Electronic Circuit

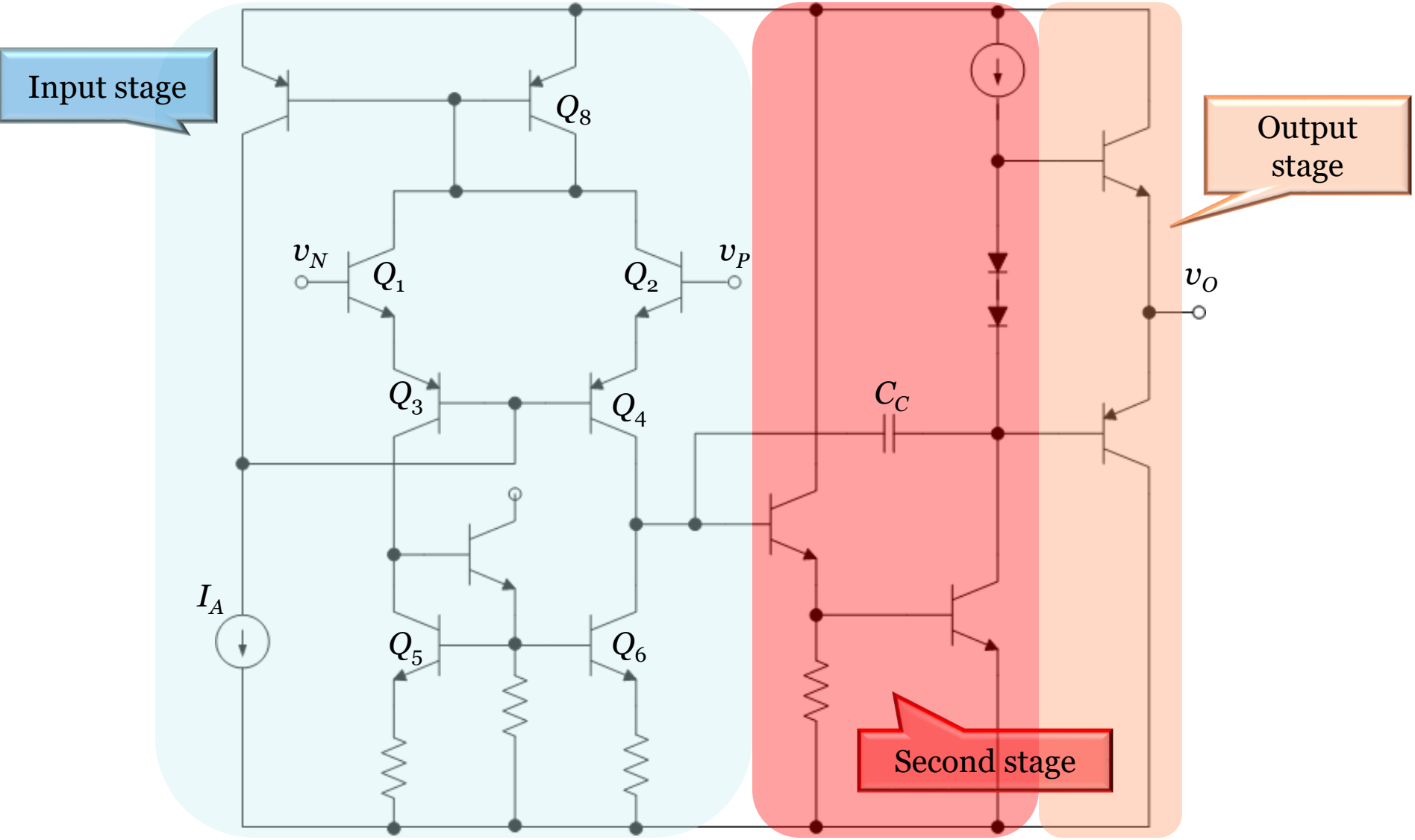
Lab1 lecture

Op Amp Limitations

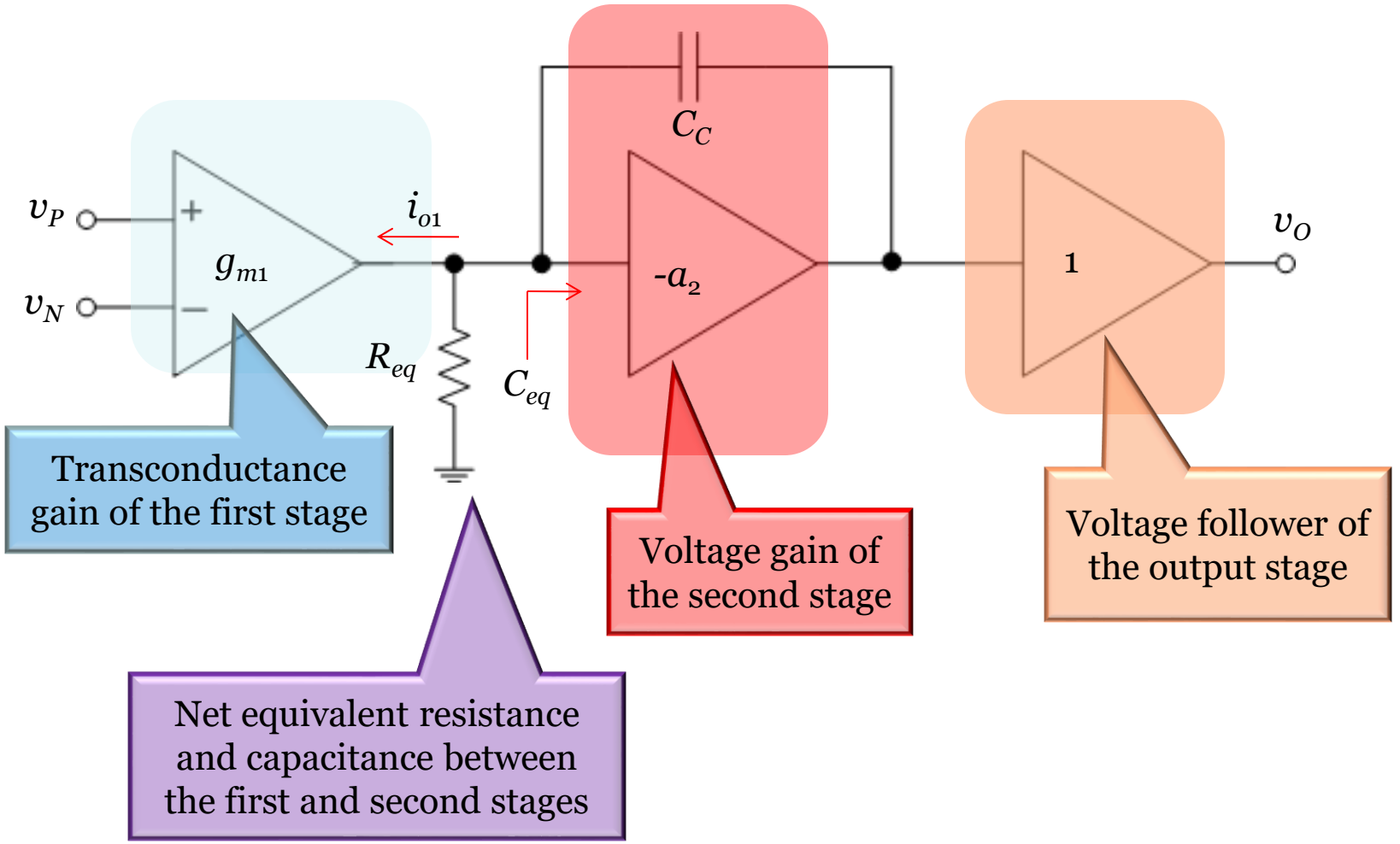
In general, under moderate frequencies, and dc gains, there is a remarkable agreement between actual behaviors of op amps and ideal op amp model.
Increasing gain or frequency, op amps are accompanied by a progression degradation.
Study op amps' limitation, predict their effect on circuit performance, and to find cures

Open-loop Response

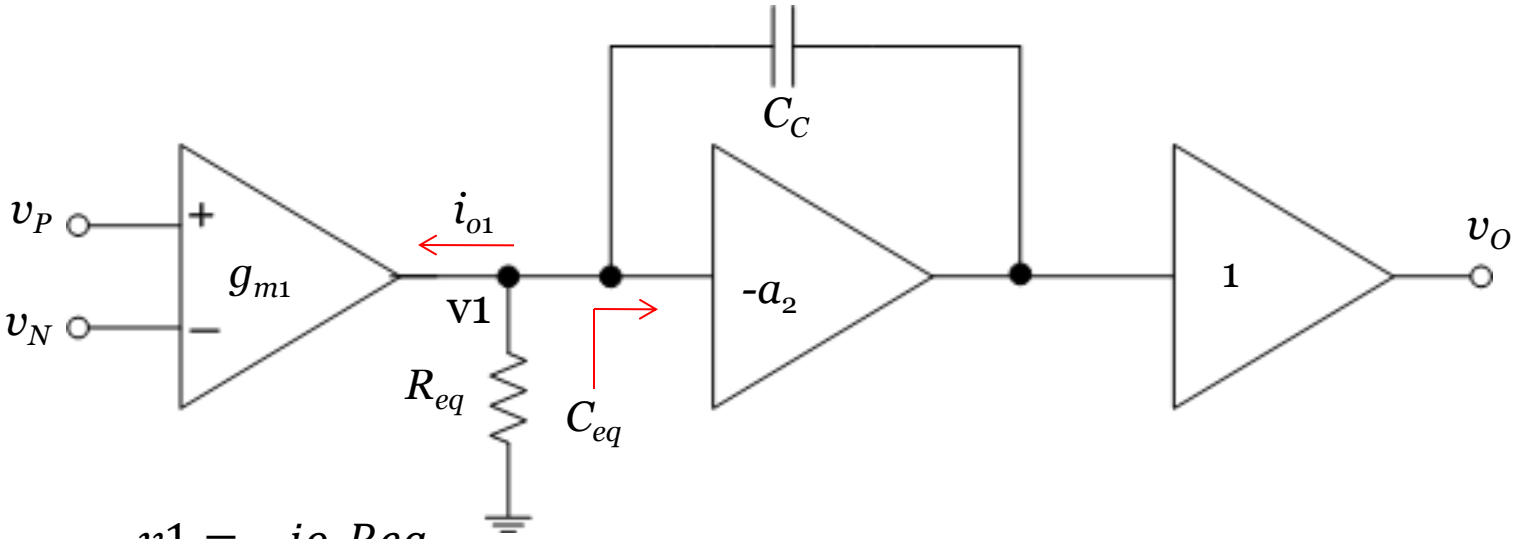
741 Op Amp Circuit Diagram



Simplified Op Amp Block Diagram



Simplified Op Amp Block Diagram



$$v1 = -io_1 Req$$
$$= (\mathrm{Vp} - \mathrm{Vn})\mathrm{gm1}(-\mathrm{Req})$$
$$= -\mathrm{Vi}\ \mathrm{gm1Req}$$

$$Vo = V1(-a2)(1)$$
$$= Vi\ gm1\ Req\ a2$$

Low frequency: C_c is open circuit, $\therefore a_0 = g_{m1} R_{eq} a_2$

For the 741 op amp, g_{m1} = 189 μA/V; R_{eq} = 1.95 MΩ, and a_2 = 544 V/V, those yields Find ao.

DC open-loop gain = 200,000 V/V.

Simplified op amp block diagram

$$H(jw) = \frac{Req}{1 + \frac{jw}{wb}}$$

$= ½$ where wb=1/(2πReq Ceq)

While increasing frequency, the equivalent impedance would decrease, gain starts to roll off at the frequency, f_b, when it is at half power,

$$\left|Z_{C_{eq}}\right| = R_{eq} \text{ or } 1/2\pi f_b C_{eq} = R_{eq}$$

This frequency, called dominant-pole frequency, f_b,

$$f_b = \frac{1}{2\pi R_{eq} C_{eq}}$$

For the 741 op amp, dominant-pole frequency = 5 Hz to make sure higher order root gain below unity. Find C_{eq}.

$$\therefore C_{eq} = \frac{1}{2\pi R_{eq}(5\ Hz)} = 16.3\ nF$$

Huge

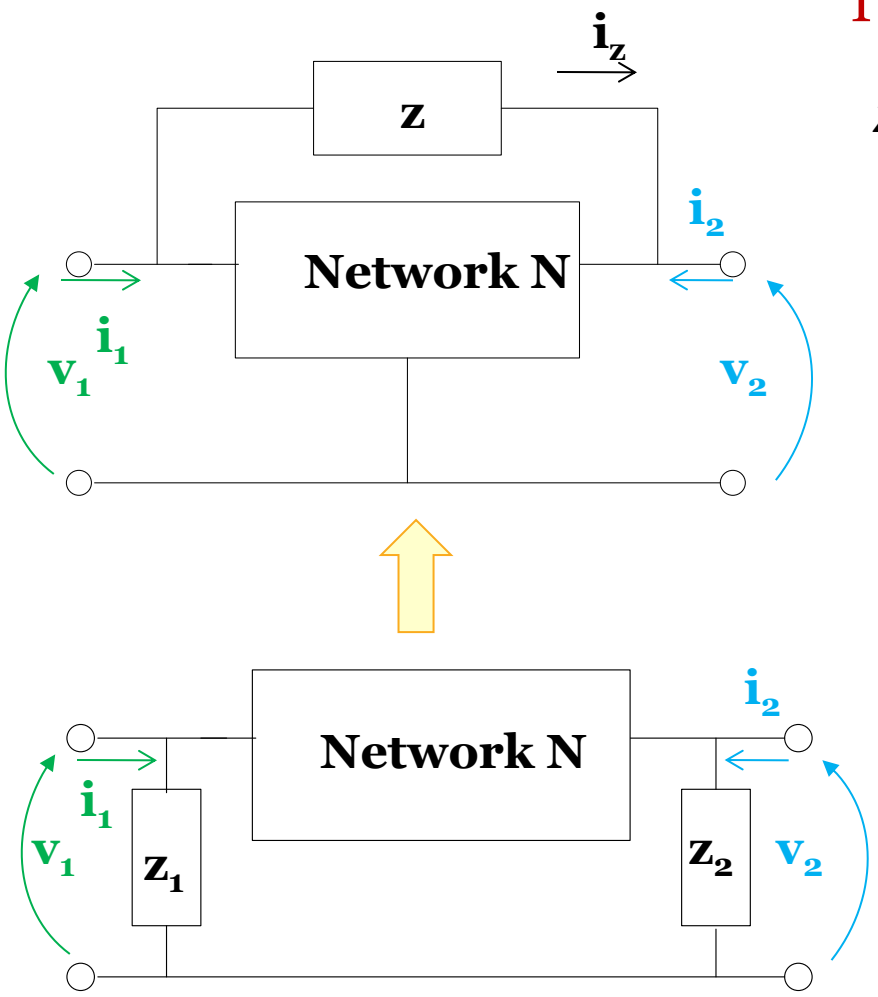
Simplified op amp block diagram

To achieve this high capacitance, large device cross section area is required,

$$C_{eq} = \frac{\varepsilon A}{d}$$

In order to avoid high cross section area, an additional capacitor Cc is added in parallel to amplifier to provide an acceptable capacitance to start with, and its equivalent value can be realize using Miller's Theorem.

Miller's theorem



Known voltage gain A_v $\quad z_1 = \dfrac{z}{1 - A_v}$

This is identical to

$Z1 = \dfrac{1}{sC_{eq}}$; $Z = \dfrac{1}{sCc}$ $\qquad z_2 = \dfrac{z}{1 - \dfrac{1}{A_v}}$

$$Z_1 = \frac{1}{sC_{eq}} = \frac{Z}{1 - Av} = \frac{\frac{1}{sC_c}}{1 + a_2} \quad (A)$$

From (A), $\dfrac{1}{sC_{eq}} = \dfrac{1}{sC_c(1 + a_2)}$ (B)

C_{eq} can be realized using Miller's effect:

From (B) and before:

$C_{eq} = 16.3nF \; and \; a_2 = -544$

$C_{eq} = C_c(1 + a_2) \Rightarrow C_c = 30pF$

Expression for the Open-loop Gain

f_b is chosen low enough that other op amp internal frequency poles and zeros are insignificant.

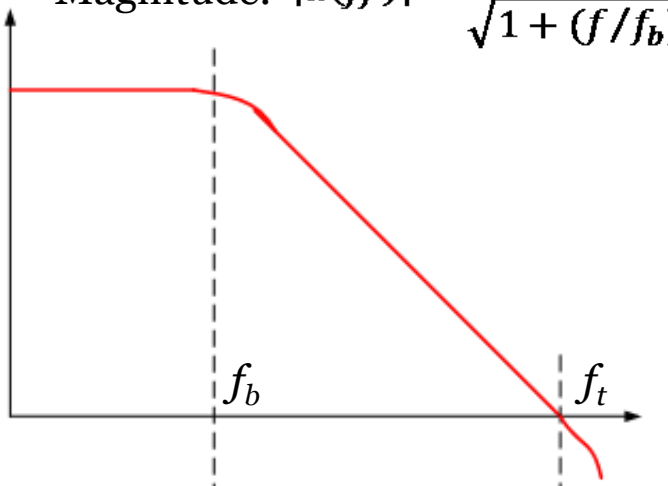
Hence, 741 is approximated with dominant pole, f_b, -3dB frequency, open loop frequency:

$$a(jf) = \frac{a_0}{1 + jf/f_b}$$

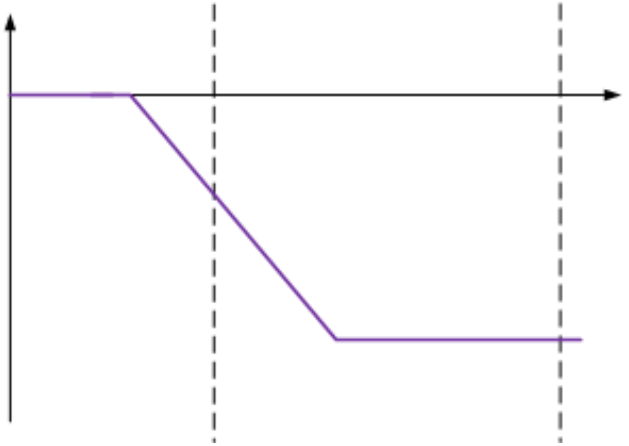
$a_0 \equiv$ Dc open-loop gain
$f_b \equiv$ *open-loop bandwidth*

For the 741 op amp, dominant-pole frequency, f_b = 5 Hz to make sure higher order root gain below unity.

Magnitude: $|a(jf)| = \frac{a_0}{\sqrt{1 + (f/f_b)^2}}$



Phase: $\angle a(jf) = -\tan^{-1}(f/f_b)$



Expression for the Open-loop Gain

f_t is unity-gain frequency or transition frequency (from amplification to attenuation)

$$1 = \frac{a_0}{\sqrt{1 + (f_t/f_b)^2}}$$

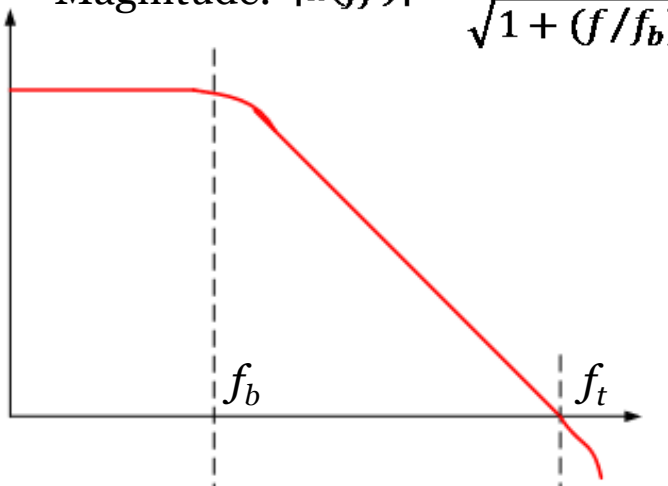
$$\Rightarrow f_t = \sqrt{a_0^2 - 1}\, f_b \cong a_0 f_b = GBP$$

$a_0 >> 1$

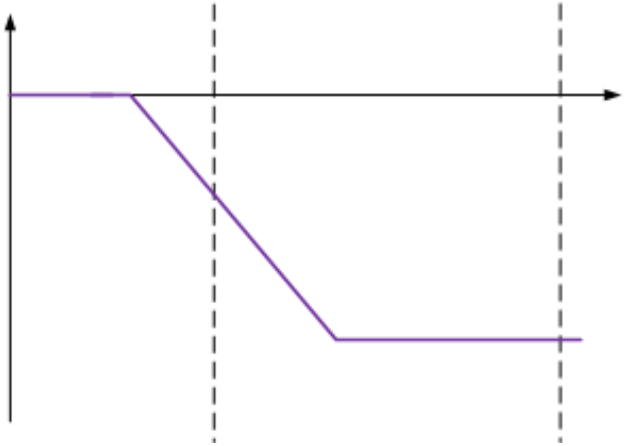
GBP ≡ Gain bandwidth product = f_t.
Every point on -20db?decade line meets *GBP* = f_t.

For the 741 op amp, unity-gain frequency = 1 MHz.

Magnitude: $|a(jf)| = \frac{a_0}{\sqrt{1 + (f/f_b)^2}}$



Phase: $\angle a(jf) = -\tan^{-1}(f/f_b)$



Expression for the Open-loop Gain

At high frequency, impedance dominates by capacitance, Z_{Cc}:

$$v_O = 1 \times Z_{C_c} \times i_{O1} = \frac{1}{2\pi f C_c} g_{m1}(v_P - v_N)$$

$$\therefore \frac{v_O}{v_d} = \frac{v_O}{v_P - v_N} = a = \frac{g_{m1}}{2\pi f C_c}$$

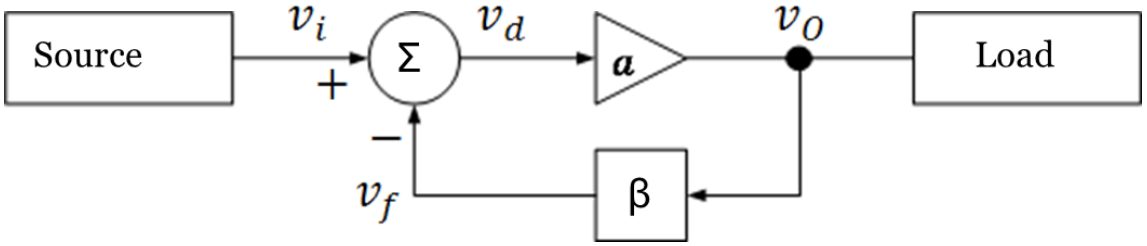
Note:
a∝ 1/f

with $a = 1$, $f_t = \frac{g_{m1}}{2\pi C_c} \Rightarrow \omega_t = \frac{g_{m1}}{C_c}$

For the 741 op amp, $g_{m1} = I_A/4V_T$, C_c = 30 pF; with unity-gain frequency = 1 MHz. Find I_A.

For the 741 op amp, $g_{m1} = I_A/4V_T$, C_c = 30 pF; therefore, pick I_A = 19.6 μA, unity-gain frequency = 1 MHz.

Closed-loop Response



Closed-loop Response

Open-loop gain: $a(jf) = \dfrac{a_0}{1 + j\,f/f_b}$

From the feedback loop:

$$A = \frac{v_O}{v_i} = \frac{a}{1 + a\beta}$$

With feedback circuits, the feedback factor: $\beta(jf)$

Loop gain: $T(jf) = a(jf)\beta(jf)$

Closed-loop gain: $A_{CL}(jf) = \dfrac{a}{1 + a\beta} = \dfrac{a(jf)}{1 + T(jf)}$

With a very large loop gain, then $A_{CL}(jf) = \dfrac{1}{\beta(jf)}$

Feedback network control

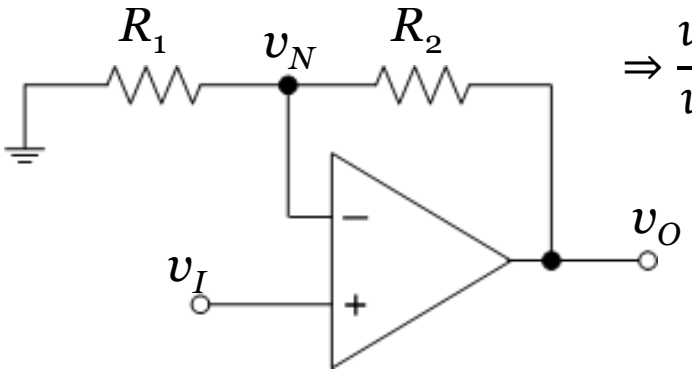
Next slide for details

Open loop frequency response: $a(jf) = \frac{a_0}{1 + jf/f_b}$

Closed-loop Response

Non-inverting op amp:

$$v_o = a(v_p - vn) = a(vI - \frac{R_1}{R_1 + R_2} v_o)$$



$$\Rightarrow \frac{v_o}{v_I} = \left(1 + \frac{R_2}{R_1}\right) \frac{1}{1 + \frac{1 + \frac{R_2}{R_1}}{a}} = \left(1 + \frac{R_2}{R_1}\right) \frac{1}{1 + \frac{1 + \frac{R_2}{R_1}}{\frac{a_o}{1 + \frac{jf}{f_b}}}} = \mathrm{A}(jf)$$

$$\beta_o = \frac{VI}{Vo} = \frac{R_1}{R_1 + R_2}$$

Close loop frequency response: $A(jf) = Ao/(1 + \frac{jf}{f_B})$

and $\frac{1}{\beta_o} = 1 + \frac{R_2}{R_1}$

Closed-loop gain

$$A_o = \left(1 + \frac{R_2}{R_1}\right)$$

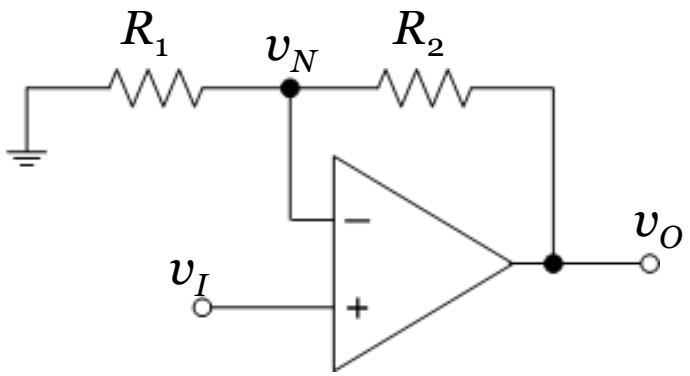
$$f_B = closed\ loop\ flat\ band\ gain = f_b \left(1 + a_o \frac{R1}{R1 + R2}\right)$$

Closed-loop Response

$$f_B = f_b\left(1 + a_o\frac{R1}{R1+R2}\right)$$

$$f_B = \text{closed loop flat band gain} \approx f_b\left(a_o\frac{R1}{R1+R2}\right)$$

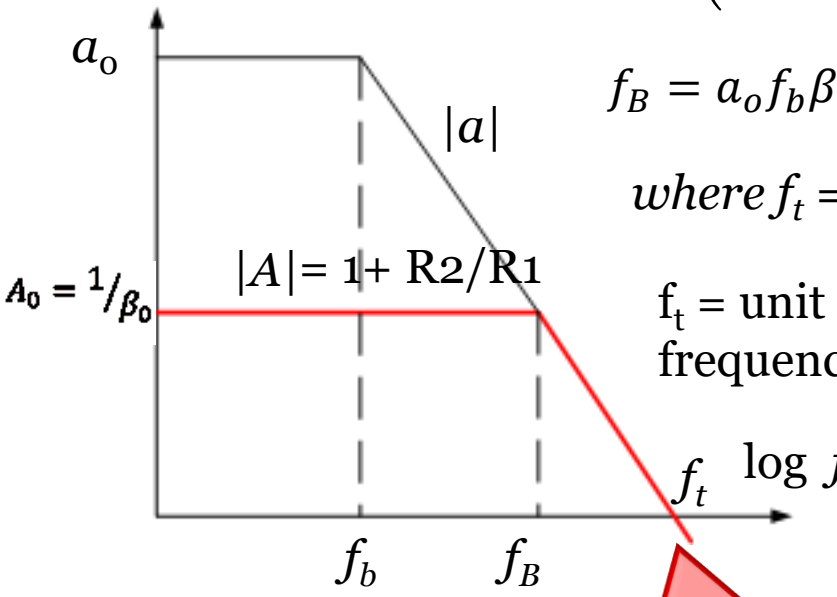
Non-inverting op amp:



$$\beta_o = \frac{VI}{Vo} = \frac{R_1}{R_1 + R_2}$$

and $$\frac{1}{\beta_o} = 1 + \frac{R_2}{R_1}$$

Closed-loop gain



$$f_B = a_o f_b \beta = \beta f_t$$

$where\ f_t = a_o f_b$

f_t = unit gain frequency

Recall $GBP = f_t$

Bode plot of T is difference between the individual plots of a and $(1/\beta)$ *next slide*.

Closed-loop Response

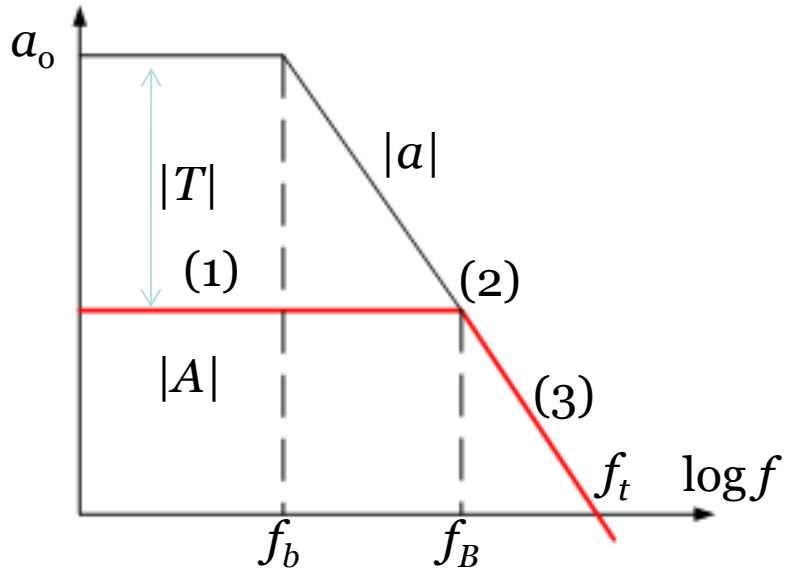
Non-inverting op amp:

$$A_{CL}(jf) = \frac{a(jf)}{1 + a(jf)\beta(jf)} = \frac{1}{\beta(jf)} \frac{T(jf)}{1 + T(jf)}$$

From

Closed-loop gain is offset by error function from ideal case

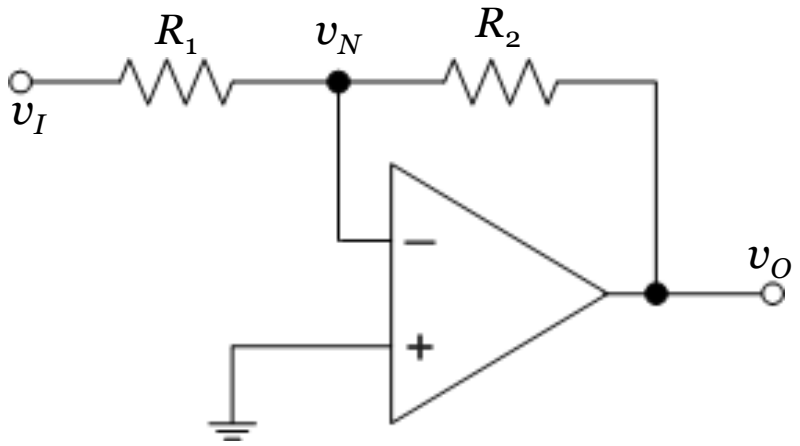
1. At low frequency, $T >> 1$, error is small and $A \rightarrow A_{ideal}$
2. At f_B, $A = A_{ideal}/(1+j(f/f_B)) = A_{ideal}/(1+j)$
3. At high frequency, $T << 1$, $A \rightarrow a$



Why Not Use Inverting Amplifiers?

Closed-loop Response

Inverting Op Amp:



$$A(jf) = A_0 \times \frac{1}{1 + jf/f_B}$$

$$A_0 \cong -\frac{R_2}{R_1} \quad f_B \cong \beta f_t$$

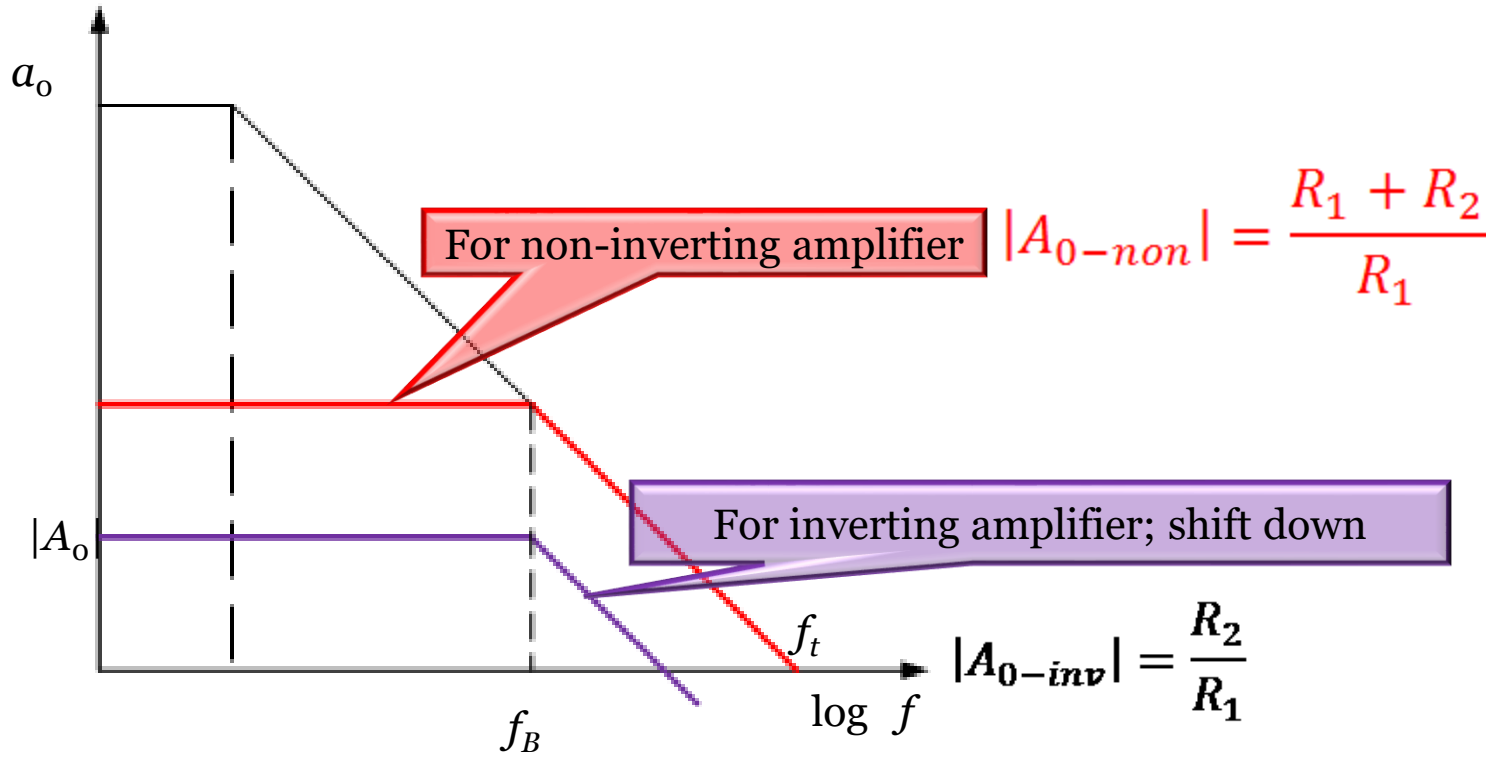
Where: $\beta = \frac{R_1}{R_1 + R_2}$ = Same value as in non-inverting case

Therefore,

$$GBP = |A_0| \times f_B = \frac{R_2}{R_1} \times \frac{f_t R_1}{R_1 + R_2} \Rightarrow GBP = \frac{R_2}{R_1 + R_2} f_t = (1 - \beta) f_t$$

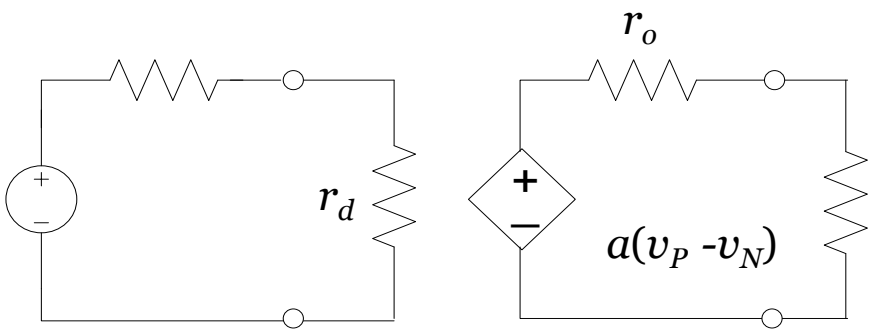
Closed-loop Response

Inverting Op Amp:



Input Impedance

Op Amp Input/Output Impedance with Feedback



r_o and r_d are open-loop output/input impedances
We normally make the assumption that $r_o \rightarrow 0$; $r_d \rightarrow \infty$.
But in reality, r_o and r_d are not ideal.

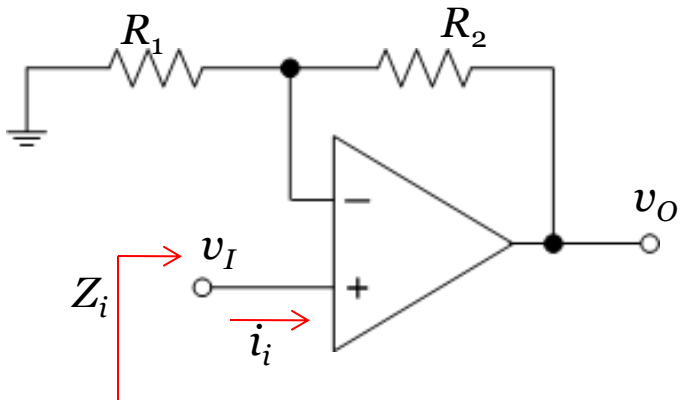
With voltage-sampling/voltage-series feedback, closed-loop R_i and R_o are both improved!

Results: $R_i = (1+a\beta)r_d$; $R_o = r_o/(1+a\beta)$

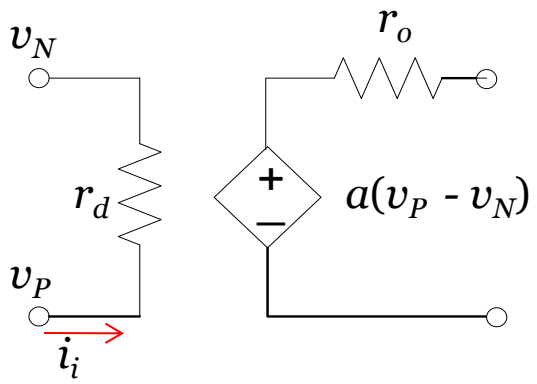
R_i increases; R_o decreases

Input impedance

Use non-inverting op amp as an example:







If $v_N = 0$; then $Z_i = r_d$

Case of no feedback

$$i_i = \frac{v_P - v_N}{r_d}$$

$$v_N = v_O \times \frac{R_1}{R_1 + R_2} = \beta v_O$$

Case of feedback

Input Impedance

Use non-inverting op amp as an example:

$$i_i = \frac{v_P - v_N}{r_d} \qquad \text{(C)}$$

Also: $v_o = a(v_P - v_N)$ (A)

$$v_N = v_O \times \frac{R_1}{R_1 + R_2} = \beta v_O \qquad \text{(B)}$$

Substituting (A) into (B), this yields $v_N = a\beta(v_P - v_N)$.

Rearranging and collecting,

$$v_N(1 + a\beta) = a\beta v_P \quad \Rightarrow v_N = v_P \frac{a\beta}{(1 + a\beta)} \qquad \text{(D)}$$

Substituting (D) into i_i equation (C),

$$i_i = \frac{v_P - v_N}{r_d} \quad = \frac{1}{r_d}\left[v_P - v_P \frac{a\beta}{(1 + a\beta)}\right] = \frac{v_P}{(1 + a\beta) r_d}$$

Input impedance

Use non-inverting op amp as an example:

$$i_i = \frac{v_P}{(1 + a\beta)r_d}$$

So that $Z_i = \frac{v_P}{i_i} = (1 + a\beta)r_d$ (Zi increase with dc gain, a)

If the op amp has a single pole at f_b, then

$$a = \frac{a_0}{1 + jf/f_b} \quad \text{where } f_t = a_0 f_b \qquad \beta = constant = \frac{R1}{R1 + R2}$$

$$Z_i = \left(1 + \frac{a_0\beta}{1 + jf/f_b}\right) r_d \quad \Rightarrow Z_i = \frac{1 + jf/[f_b(1 + a_0\beta)]}{1 + jf/f_b}(1 + a_0\beta)r_d$$

Frequency dependence

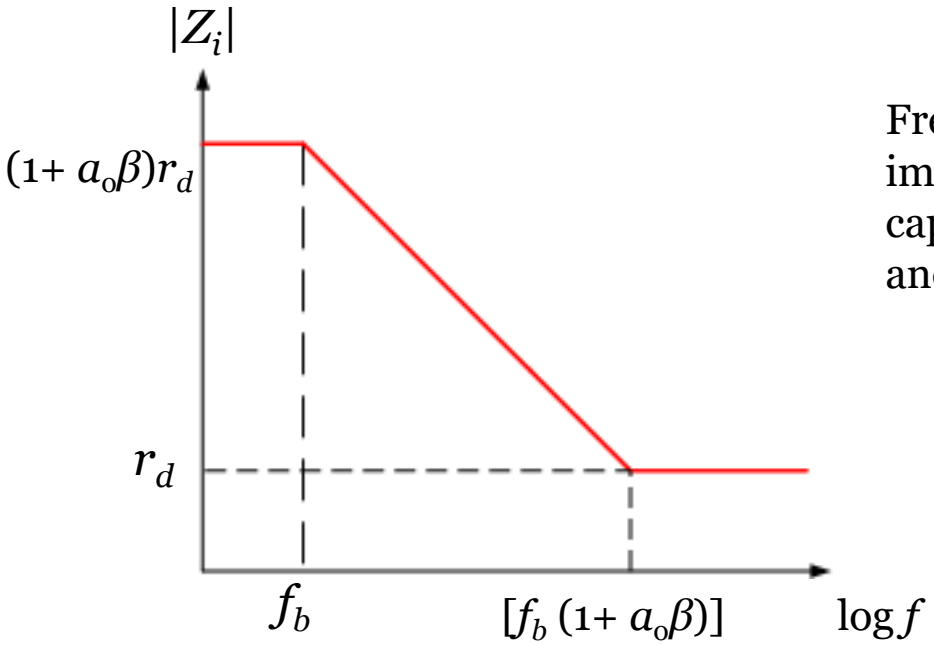
DC portion

Input Impedance

Use non-inverting op amp as an example:

$$Z_i = (1 + a_0\beta) r_d \frac{1 + jf/[f_b(1 + a_0\beta)]}{1 + jf/f_b}$$

Pole: f_b; zero: $f_b(1+a_o\beta)$



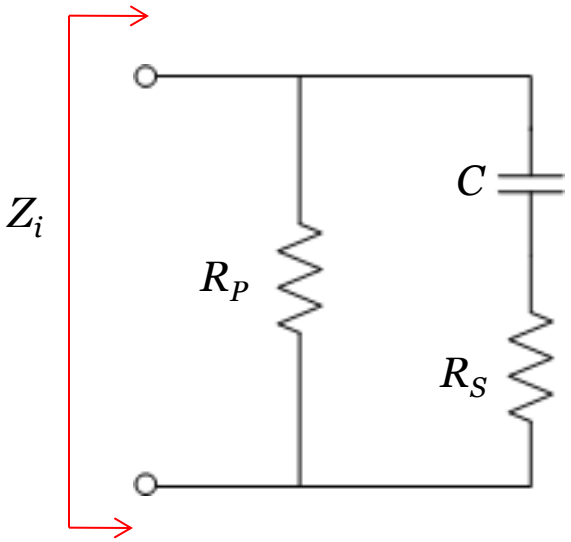
Frequency increasing, input impedance decreasing → like a capacitor ⇒ capacitive between f_b and $f_b(1+a_o\beta)$.

Input Impedance

$$Z_i = R_P//(1/sC + R_s) = R_P//(\frac{1+sR_sC}{sC})$$

$$= \frac{R_P \times \frac{1+sR_sC}{sC}}{R_P + \frac{1+sR_sC}{sC}} = \frac{R_P \times (1+sR_sC)}{(1+s(R_P+R_s)C)} = R_P \frac{1-\frac{s}{Z}}{1-\frac{s}{P}}$$

At low frequency, $\boxed{Zi = Rp \cong (1 + a_o\beta)r_d}$



At high frequency, capacitor as a short circuit,

$Z_i \cong r_d$ (see previous plot)

$R_P//R_s = r_d$; but $R_P >> r_d$, so $\boxed{Rs \cong rd}$

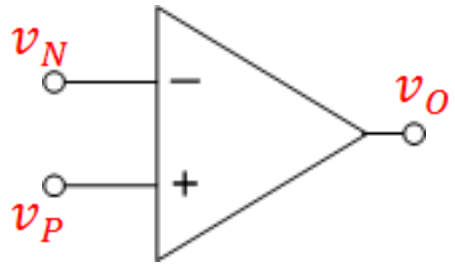
$$f_c = [fb(1 + ao\beta)] = \frac{1}{2\pi RSC}$$

$$\boxed{C = \frac{1}{2\pi r_d[f_b(1+a_0\beta)]}}$$

Input Offset Voltage

Input offset voltage

Ideal op amp



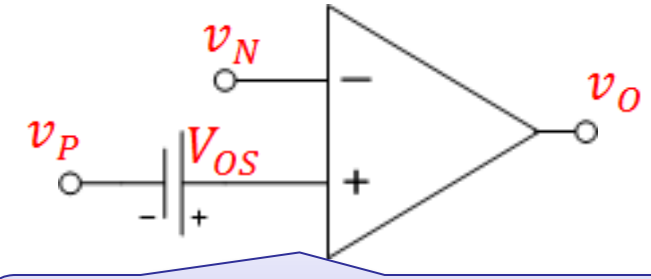
$$v_O = a(v_P - v_N)$$

$$\therefore v_O = 0 \ (\text{with} v_N = v_P)$$

Op amp with VOS

$$Vos = VT \ ln \frac{I_{s1}}{I_{s2}} \frac{I_{s4}}{I_{s3}}$$

$$I_s = \frac{qD_B}{N_B} \ {n_i}^2(T) \frac{A_E}{W_B}$$



With input referred DC offset voltage VOS, which can be in series with either v_P or v_N *(mismatch of transistor pair)*

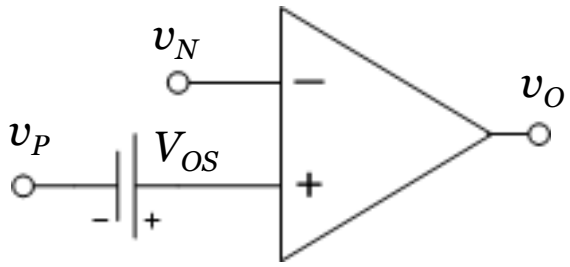
$$v_O = a(v_P + V_{OS} - v_N)$$

$$\therefore v_O \neq 0 \ (\text{with} v_N = v_P)$$

Note: small V_{OS} can saturate op amp output in open loop case

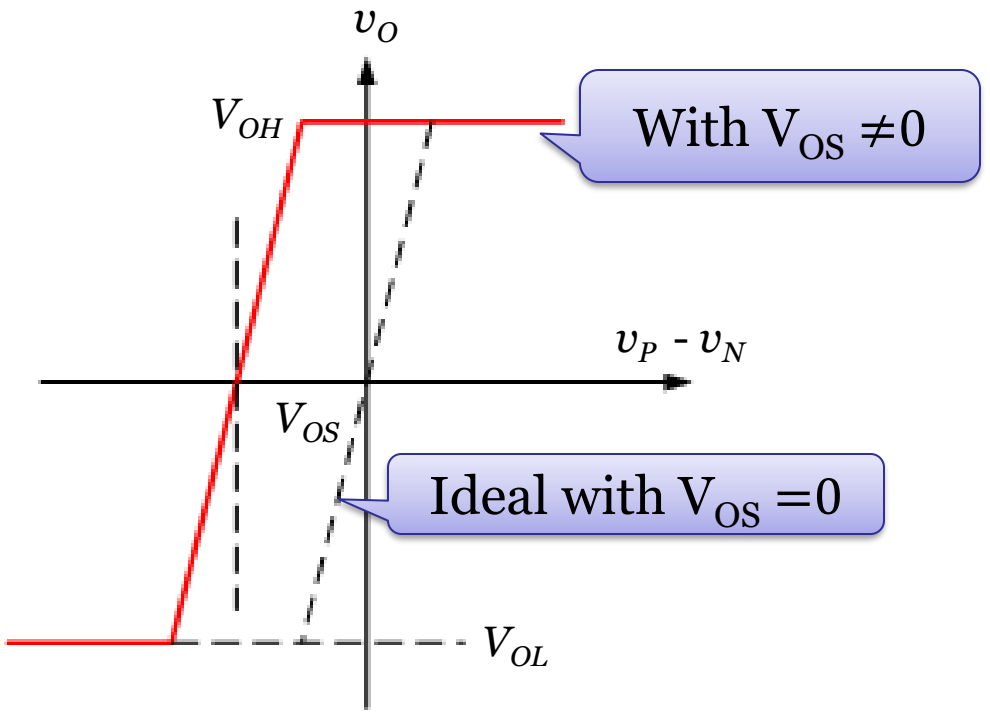
Input Offset Voltage

Op amp with VOS



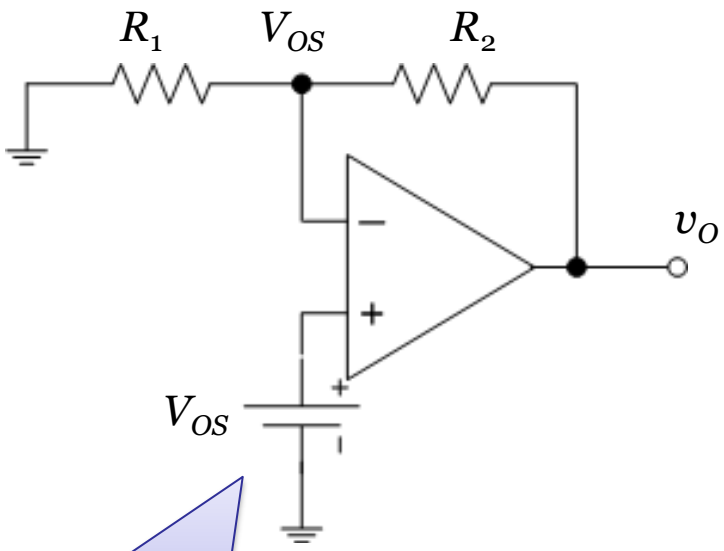
$$v_O = 0 \Rightarrow v_N = (v_P + V_{OS})$$

$$\therefore v_P - v_N \neq 0 (\text{with } V_{OS})$$



VTC is shifted by V_{OS}

Errors Caused by V_{OS}



As non-inverting configuration with input V_{OS}

Find Vo in terms of the circuit parameters.

$v_N = V_{OS}$

Applying KCL at v_N,

$$\frac{v_N}{R_1} + \frac{v_N - v_O}{R_2} = 0$$

$$v_N\left(\frac{1}{R_1} + \frac{1}{R_2}\right) = \frac{v_O}{R_2}$$

$$\therefore v_O = E_O = R_2 V_{OS}\left(\frac{1}{R_1} + \frac{1}{R_2}\right) = (1 + \frac{R_2}{R_1})V_{OS}$$

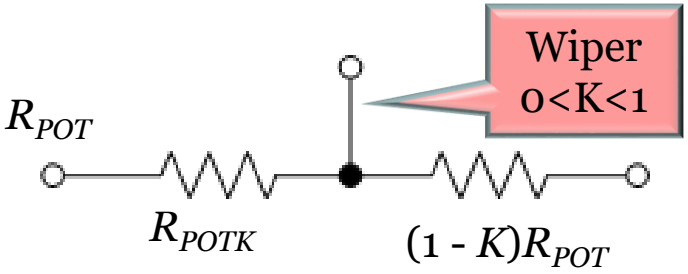
Offset error referred to the output

dc noise gain

Input Offset-error Compensation

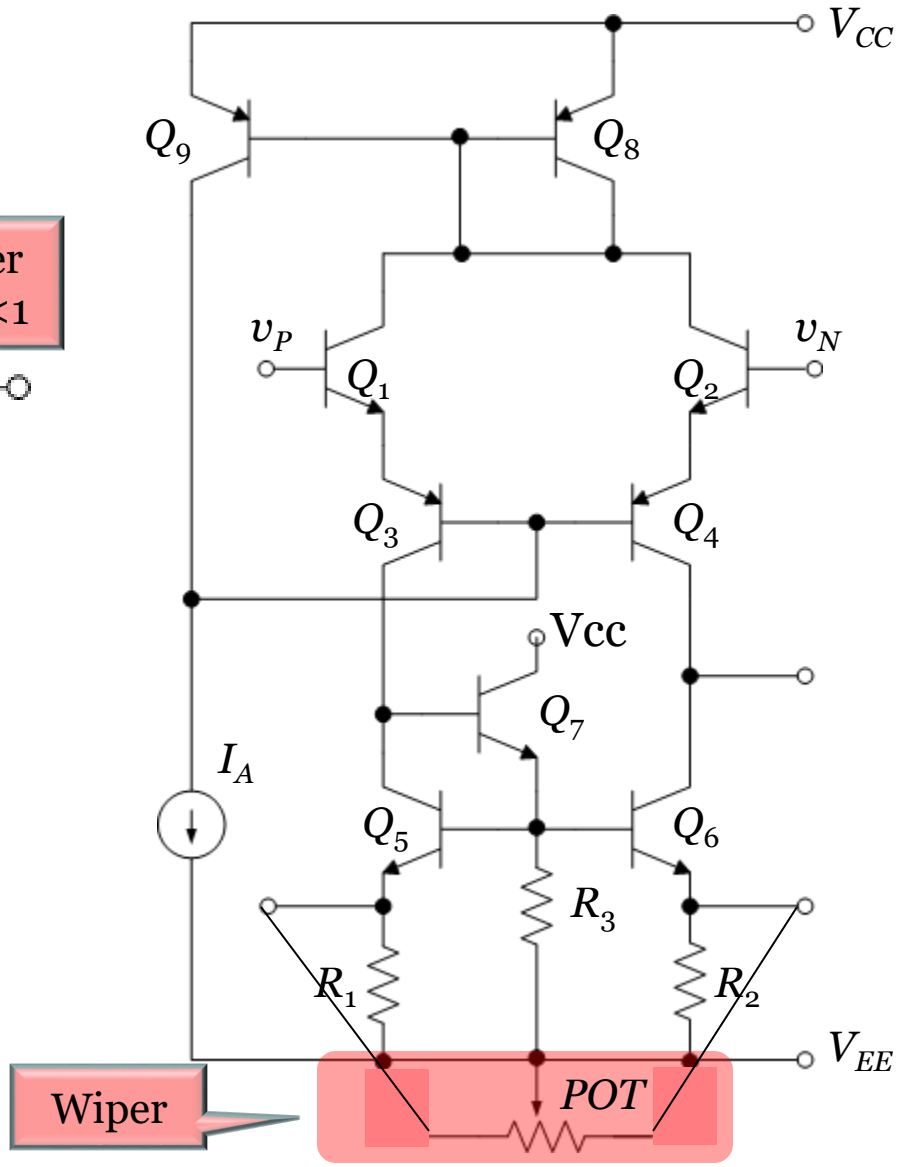
Internal Offset Nulling

Use 741 as an example:



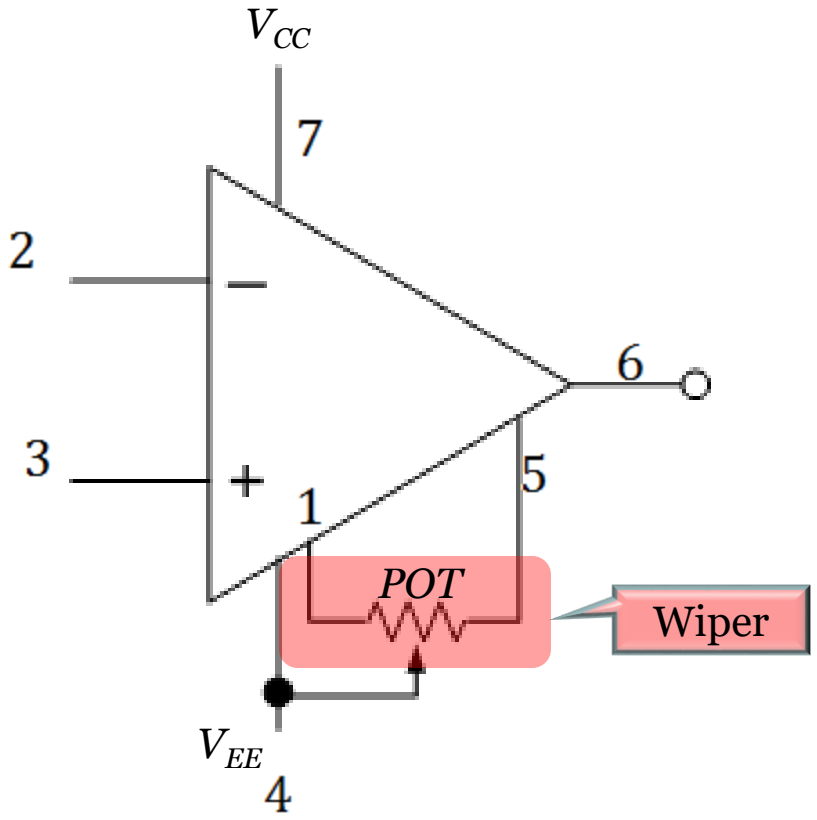
Key points:

1. With v_I=0V, adjust wiper for v_O=0V
2. Size of the POT is about 10X larger than R_1 and R_2
3. Read the Data sheet (1 Kohm)



Internal Offset Nulling

Use 741 DIP-8 package as an example:

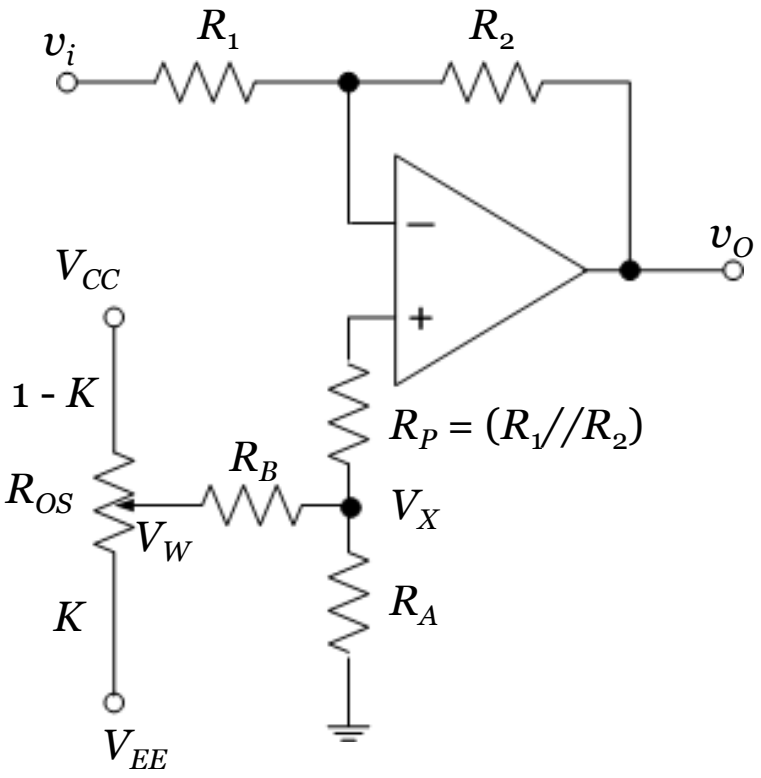


External Offset Nulling

Some op amps provide no pins for offset error compensation, so we must compensate them through external voltages or currents placed at input to op amps.

The common op amps without internal offset nulling are quad-pack packages, such as LM 348.

External Offset Nulling



Choose $R_A << R_P$ to keep canceling I_B effective and not disturb the I_B compensation

Choose $R_B >> R_{OS}$ to reduce loading on the R_{OS} voltage divider.

Assuming very small current pulled to R_B, since $R_B >> R_{OS}$

$$V_w = \left(\frac{K}{K + 1 - K}\right)(V_{CC} - VEE) + VEE$$

$$\boldsymbol{V_W = (KV_{CC} + (1 - K)V_{EE})}$$

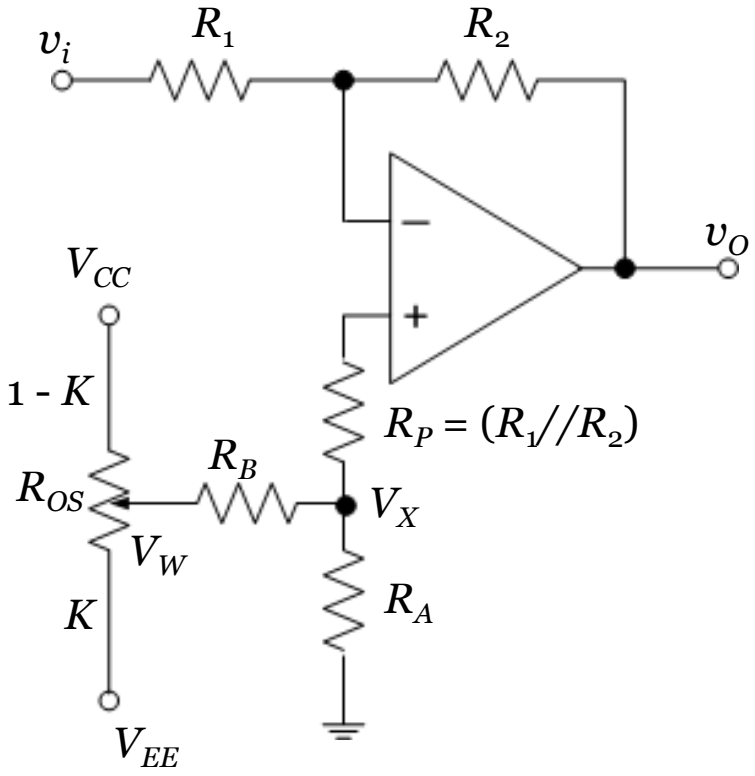
and $$\boldsymbol{V_X = V_W \frac{R_A}{R_A + R_B} \cong V_W \frac{R_A}{R_B}}$$

By adjusting V_X, we can cancel the offset of V_{OS}.

External Offset Nulling

Example: A 741 op amp is to be used in the circuit to yield A = -5 V/V and R_1 = 30 kΩ. Specify suitable resistance.

For μA741 op amp:
V_{OS} = 6 mV;
I_{OS} = 200 nA (worst case);
I_B = 500 nA



External Offset Nulling

$$V_X = V_W \frac{R_A}{R_A + R_B} \cong V_W \frac{R_A}{R_B}$$

Solution

R_1 = 30 kΩ; R_2 = 150 kΩ

Choose: $R_P = R_1 // R_2$ = 25 kΩ → 24 kΩ

For μA741 op amp, V_{OS} = 6 mV; I_{OS} = 200 nA (worst case);
I_B = 500 nA

Therefore, the worst case,

Ei = Vos + Rp Ios = 6mV + 24k x 200nA = 11 mV

Total offset input error is in the range (-11 mV, 11 mV).

External Offset Nulling

$$V_X = V_W \frac{R_A}{R_A + R_B} \cong V_W \frac{R_A}{R_B}$$

Example: A 741 op amp is to be used in the circuit to yield A_s = -5 V/V and R_i = 30 kΩ. Specify suitable resistance.

Solution

Choose: -15 mV < V_X <15 mV to cover E_i range, and use V_{CC} = 15 V and V_{EE} = -15 V, therefore, pick V_W = 15 V (or -15V) to cover the range,

Need to pick: $R_A/R_B = 10^{-3}$ and $R_P = 24\ k\Omega$

Set: $R_A = 1\ k\Omega$ $R_B = 1\ M\Omega$